

Mohammad Yusuf

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PROFILE

Bioscience graduate and incoming M.Sc. Bioinformatics student with hands-on experience in computational biology, genomic data analysis, and machine learning-driven scientific workflows. Proficient in Python, R, and Linux, with demonstrated expertise in sequence analysis, molecular docking, and biological network analysis. Proven leadership founding and scaling a student research community, combined with over a year of experience structuring biological and technical data for AI model training. Seeking a research or industry role to apply computational biology and bioinformatics skills toward impactful scientific discovery.

EXPERIENCE

- Bioinformatics Intern — Bioture**

Jul 2025 – Aug 2025

 - Utilized R programming for statistical analysis and visualization of complex biological data, processing datasets exceeding 2 GB to identify differential gene expression patterns.
 - Operated within a Linux environment executing standard RNA-seq pipelines across 10+ genomic sequences, automating routine scripts to accelerate data processing and cut manual analysis time.
 - Performed sequence analysis and functional annotation on 15+ gene sequences, accurately interpreting genomic data to identify gene functions and biological relevance.
 - Conducted molecular docking studies on 8+ protein-ligand complexes using AutoDock Vina, successfully identifying the top 5 high-affinity binding candidates for downstream analysis.
 - Constructed and analyzed gene interaction networks spanning 50+ nodes with Cytoscape, visualizing key biological pathways to support research interpretation.
- Founder & Lead, Genomics and Data Science Club — Jamia Millia Islamia**

Jan 2025 – Jul 2025

 - Founded and scaled the university's first Genomics and Data Science Club, growing active membership to 50+ students within 6 months.
 - Directed 2 technical workshops covering Python, R, and machine learning concepts, training an average of 70 attendees per session.
 - Facilitated 3 collaborative student projects in computational biology, fostering a peer-driven research and learning community.
- AI Trainer (Biology & Basic Programming) — Outlier**

Aug 2023 – Dec 2024

 - Evaluated and refined 80+ AI-generated biological prompts per week over a 17-month engagement, ensuring 95%+ scientific accuracy and structural clarity for machine learning deployment.
 - Developed and delivered training materials on fundamental biology and programming concepts, supporting continuous improvement across multiple AI model iterations.
 - Structured and annotated 100+ complex biological data points into machine-readable formats, directly improving the model's educational output quality for foundational science concepts.

EDUCATION

- Master of Science, Bioinformatics — Department of Computer Science**

Jamia Millia Islamia | Delhi, India

Expected: 2028
- Bachelor of Science, Bioscience**

Jamia Millia Islamia | Delhi, India

Expected Graduation: Jun 2026
- Relevant Coursework: Bioinformatics, Molecular Biology, Developmental Biology, Biotechnology and Human Welfare, Animal Physiology, Plant Physiology, Biostatistics, Chemistry.

Key Lab Skills: SDS-PAGE, ELISA, PCR, Spectrophotometry, Cell Extraction, Gel Electrophoresis.

ENTRANCE EXAMINATIONS & ACADEMIC ACHIEVEMENTS

- GAT-B (Graduate Aptitude Test in Biotechnology) — All India Rank 99
- M.Sc. Bioinformatics Entrance Examination, Jamia Millia Islamia — Rank 2
- M.Sc. Biophysics Entrance Examination, Jamia Millia Islamia — Rank 2
- M.Sc. Microbiology Entrance Examination, Jamia Millia Islamia — Rank 5

SKILLS & ABILITIES

Programming Languages: R, Python

Technical Skills: Linux, Molecular Docking, Sequence Alignment (BLAST), Genomic Databases (NCBI, Ensembl), Biopython, Bioconductor, Data Analysis & Visualization, Git & Version Control

Soft Skills: Leadership, Communication, Training & Mentorship, Team Collaboration, Problem-Solving

CERTIFICATIONS & WORKSHOPS

- Two-Day Hands-on Workshop on Bioinformatics and Molecular Biology Techniques | TRYST, IIT-Delhi
- Introduction to the Biology of Cancer | Coursera
- Python Basics | Coursera
- Global Outreach Student Member, American Society for Microbiology (Jun 2025 – Present)

ACTIVITIES AND INTERESTS

Mentoring: Guiding students in competitive exam preparation. | Technology: Bioinformatics applications and blockchain trends. | Finance: Trading strategies and financial markets. | Creative: Nature Photography.